

SURVEY OF AUTOMATED GUIDED VEHICLE IN JAPANESE FACTORY

Toshihiro TSUMURA

College of Engineering, University of Osaka Prefecture

Mozu, Umemachi, 4-804, Sakai, Osaka 591, Japan

ABSTRACT

Recent trends and advances of automated guidance and positioning techniques and systems are surveyed, especially good for application to automated factory of many kinds in Japan.

INTRODUCTION

For advancing the recent development of full automated manufacturing system in factory, AGV(Automated Guided Vehicle) serves as a key element of transfer or transportation system by way of automated guidance ability and automation of unmanned transporter or carrier.

Many transportation methods are proposed and practiced in the past: hoist, crane, railway cart system, fork lift truck, and automated or manual controlled carrier with wheel tire truck and so on, but in the past 5 years, automated guided wheel type vehicle is evaluated as the powerful tool for transfer and carrying in FMS.

The first AGV was prepared in 1963 by Daifuku Ltd. in Osaka Japan but this AGV technique and engineering were introduced from WEB company, U.S.A. Following the above Daifuku AGV, famous five companies are new comers in this field, such as Shinko Electric, Murata Machinery, Nippon Yusoki, Toyo Unpanki Ltd and Matsuda Kusan, all in Japan.

Now, it is well known that over 30 companies are working on this AGV field, and over 15 companies made AGV in commercial base in Japan.

Now, almost all AGV are used in factory, in Japan, but many applications are known in many field, such as for office automation, public field facilities, terminal plaza service, shopping center, container yard, restaurant service, hospital service, agricultural and earth-moving field, construction and fabrication of buildings, and so on.

Here, recent advances of the automated guidance and positioning techniques and systems are surveyed and discussed on AGV research and development, especially discussions are restricted to the factory automation field in Japan.

AUTOMATED GUIDANCE SYSTEM

(1) Electromagnetic Guidance (Electro-magnetic Leader Cable System)

This method is most popular and very famous for automated guidance method. Almost 90 per cent of AGV adopted this guidance method. Using 3-10 kHz of signal is applied to the cable and it introduces induction voltage in the magnetic coil on the floor side of vehicle. This induced signals are used for sensing the deviation of guidance vehicle on floor in factory. Principle of this method is well known, but multiple or parallel setting of cable systems are introduced for flexibility of course arrangement. Different frequency signals are applied each cable for discriminate and select the course, communication signals are applied to the cable for station to vehicle communication and checking and setting forbidden zone for collision avoidance. Fig.1 show this guidance method.

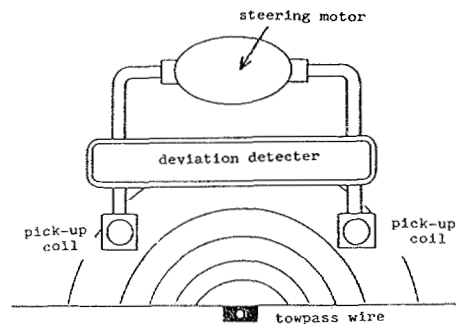


Fig. 1. Electromagnetic Guidance System

(1)(2) Reflecting Tape Guidance (Optical Tape Guidance System)

Reflecting tape (white tape, metal tape, painting with glass powder on road course) are set stucked on the route of roadway for sensing the detection of deviation from course of the ground

vehicle. Any type of light on the lower side of vehicle is reflected and this reflected light signals are sensed by photo-sensor under the vehicle.

Continuous tape setting will be better than the tape with discrete setting (for saving reflecting tape).

This tape guidance system is superior to the electromagnetic guidance system on the point of flexible setting, easy setting and good for understanding to the person walking near the vehicle. In Japan, over 5 companies recommend the tape guidance system for good application about business/office machine manufacturing company, small parts machining or making companies.

Single tape is known as good in operation but many kind of tapes are proposed, developed and recommended such as:

- a) Doubling the tape in parallel for reliability,
- b) Printing bar-code or some kind of marks are written on tape and this will be good for setting absolute data,
- c) Curved tape (many kinds of tape with variety of radius) are proposed.

Fig. 2 show principles of reflecting tape guidance method.

Photo 1 shows tape guidance system.

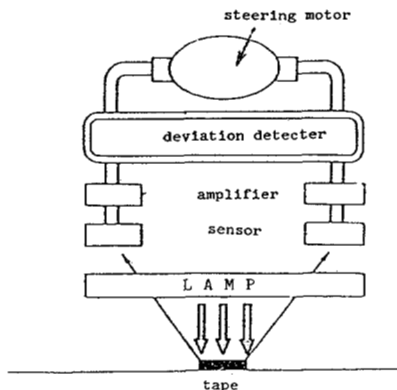


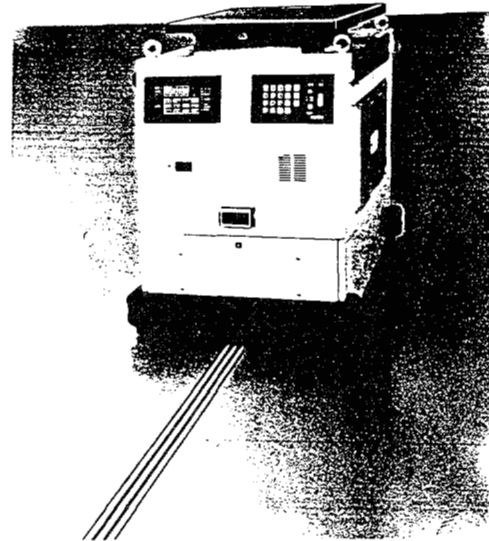
Fig. 2. Reflecting Tape Guidance System

- (3)
3) Ferrite Painted Guidance (guidance using painting course by ferrite magnet power)

Ferrite magnetic power are painted on the course of the vehicle, magnetic sensor are set on both sides of vehicle, and information sensed by magnetic sensor is introduced for deviation of vehicle from the pre-painted course. This system was proposed by new commer last October.

- (3)
4) Magnet Set Guidance

On predetermined positions magnetic sticks are hitted on the course, and magnetic sensor detect the difference between the absolute positional data and the data measured by self-sustained positional data. Therefore, this magnet on the course is applicated as checking of guidance properties of AGV.



Photo

Example of Reflecting Tape Guidance System

- (4)
5) Laser Guidance (Laser Beam Guidance System)

On the upper side of the factory or roadway, laser scanners are set and these scanners are scanning laser beam to introduce course way.

Some photo-sensing array or sensing system are set on the top or upper side of vehicle, and detected data are applied to the AGV steering circuit.

Two methods are proposed:

- a) Fixed straight line guidance system: laser beam is fixed,
- b) Arbitrally course fixing: using laser scanner, reflect laser beam by one two dimensional mirror controller.

Fig. 3 shows reflecting mirror controller for introducing arbitrary course curve.

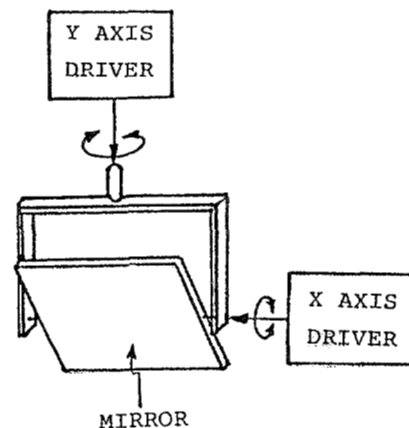


Fig. 3. Two-dimensional Mirror Controller

Now, straight line guidance system are commercially available, and this system is superior to other systems such as electromagnetic guidance and reflecting tape guidance, for carrier in super-clean room in coming years, arbitrary course guidance system by using laser will be introduced.

6) Self-Sustained Guidance System

For self-sustained guiding of vehicle, there must be measured the positional data of vehicle in continuous mode of discrete measuring with quick cycle-time. Position finding method which is good for AGV are presented in the next section. If the positional data are introduced by on-board computer, these data are compared with actual course data written on course map or memorized digital map, and if the difference is introduced or computed, this data is applied to automatic steering controller to reduce the deviation from the course.

(5)
Fig. 4 shows principle of map guidance system, in the figure, course is written on map and positional data are in computer on the vehicle and predicted positional data (after few second or after any travelling distance) are computed and mapped on the paper map. If the difference is introduced, steering motion occurred. Photo 2 shows an experimental system hardware.

If complicated curved course must be set on digital map system, curved course is approximated by piecewise linear approximation and these pieces are set by both end points by x- and y- positional data and these data are memorized in the memory of the on-board computer. Fig. 5 shows principles of memory map guidance system. This system also will be appears in commercial base.

7) Optical Recognition Guidance System

Many vision systems which will be good for vehicular or moving targets are researched, reported and developed: from the simplest mark system to the complicated and computerized large scale three dimensional vision systems are reported, but if the vision systems are used in practice, the following three methods are useful with low cost, reliable and quick processing.

a) Corner-cube laser-scanning system

Corner cubes (a kind of retroreflector) are set on the top or ceiling of the way as the reference position on course, and fan-beam laser scanner with rotary encoders are set on the top of guidance vehicle. If the scanned fan-beam hit the corner cube pair, reflected laser beam lights are received on photo sensor on the scanning system. Angles from the reflected corner cube are introduced to the on-board computer to find the deviation from the course. This guidance system also introduces heading error from the reference course. Fig. 6 shows the principle of this guidance system.

b) Bar-code Mark Recognition System

Series of bar code marks are set on course way in every few meter-spaced for guidance information to the vehicle. Video-camera systems are set on the front side and underside of the vehicle. Optical data on position of bar code patterns are processed by on-board computer to find the

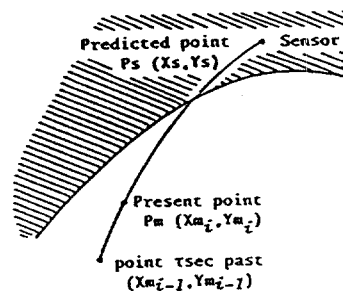


Fig. 4. Principle of Map Guidance System

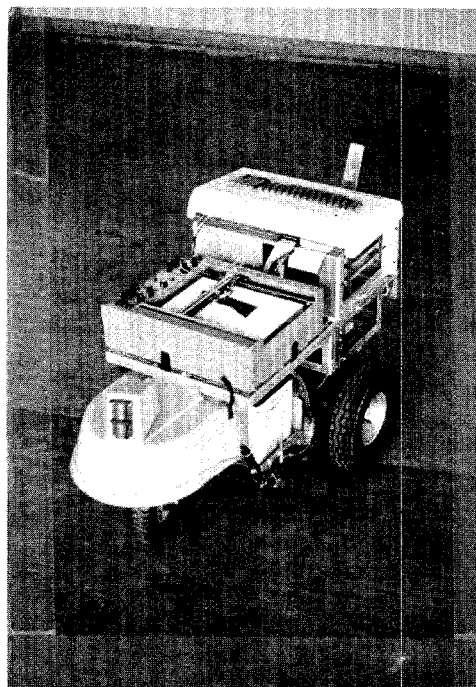


Photo 2. Example of Map Guidance System

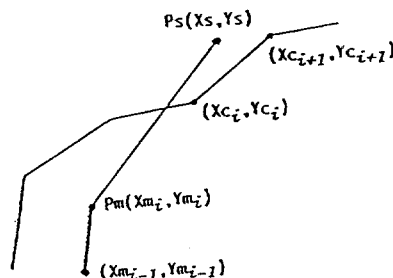


Fig. 5.

Principle of Memory-map Guidance System

positional data and guidance data. Fig.7 shows illustration of the guidance system. This system is available in Japan.

c) Spot-mark/Vision-camera System (6)

As series of spot marks, small glass balls are set on the course and these spot marks are protected and guarded by hard plastic cases. Video camera

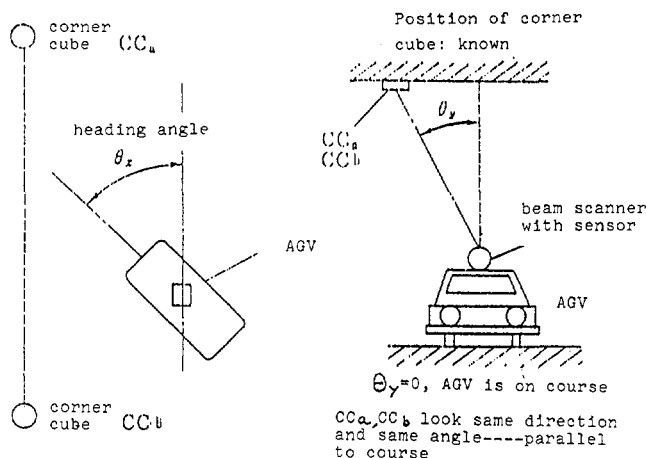


Fig. 6. Principle of Corner-cube

Laser-scanning System

systems are set on the front side of vehicle and deviation from the course route is measured at the CCD receiver and this data are applied to on board computer to find the steering information of vehicle.

Photo 3 shows spot mark and Photo 4 show the spot-mark/Vision camera system hardware.

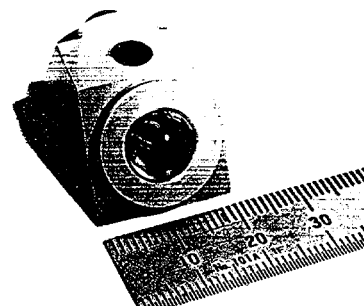


Photo 3. Example of Spot Mark on Floor (courtesy of Komatsu Ltd.)

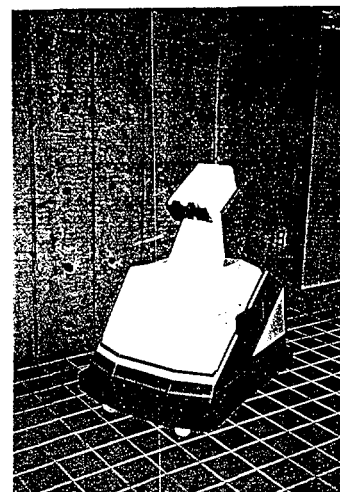


Photo 4. Example of Spot Mark/Vision-camera Guidance System (courtesy of Komatsu Ltd.)

ON BOARD POSITIONING SYSTEM

A lot of automatic position finding system is presented, discussed and manufactured in the past, but here, reasonable methods which are good for on boarding and economical as well as precise and quick processing are described.

1) Positioning using Gyroscope

In Japan, free gyro system and rate gyro systems are presented in practice. On free gyro system, precision free gyroscope is introduced for measuring direction of AGV and odometer is used for distance measurement. On board computer processed incremental positional orthogonal x-and y-data and these data are applied to teaching-playback processing element. A demonstration system shows AGV with teaching-playback type guidance.

On rate gyro system, rate gyroscope uses the principle of Coriolis force induced in gas rate sensor. This gas rate gyroscope is not so

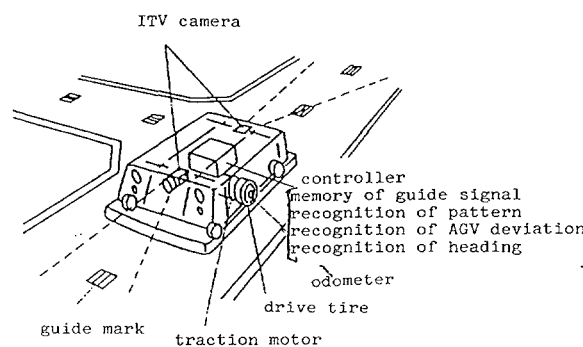


Fig. 7. Configuration of Bar-code Mark Recognition System

expensive, many companies tried to use this gyro, Fig. 8 show illustration of gas rate sensor and block diagram form operation.⁽⁹⁾

Heading rate from gas rate sensor and distance data from odometer are applied to on board micro-computer and positional data are used to steering the mechanism of AGV.

Fiber optics gyroscope is also competitive with mechanical rotary gyroscope, and many companies are interested in this rate measuring system. Magnetic compas and magnetic fluid rate sensor will be also available in coming year.

2) Position Measurement Using Odometric Data Only
Position sensing only using odometric data was proposed by Myer,⁽¹⁰⁾ and this principle is extended to the practical use, both side of wheel has precise rotational measuring system such as digital incremental encoder and at every displacement the data are processed to compute the positional and heading outputs,⁽¹¹⁾ the other proposes⁽¹²⁾ was on processing at every constant time. The latter was applied to use positional sensor for AGV with good accuracy and cheap in economy. In both methods, summing data of both wheel rotation introduces displacement and subtracting data of both wheel introduces heading change.

3) Sonic Beacon System⁽¹³⁾

Vehicle has a sonic or ultrasonic transmitter and is scanned in rotation mode, and if some reflectors are set on surface and the position data are inputted into the vehicle computer, vehicle can receive the reflected sonic waves and can read rotational angles and on board computer processed the data to find position in every fraction of second. Diverging sonic beam will introduces erroneous data, so, ultra sonic wave is preferable.

If sonic beacons are set on three positions on ground, and vehicle has sonic receiver and rotary encoder, geometric relation introduces positional data. If narrow sonic receiver is used this system may be better than the former system (vehicle have scanner and transmitter).

4) Laser Beacon System

At the predetermined position a laser scanning system is installed, and this scanning system has precision digital encoder and data transmitter to transmit the angular data to surface vehicle. Vehicle have three photo sensor to receive the rotating scanning laser beam at three points on vehicle and at that receiving time, transmitted angular data are received to process. Principle of geometric relations are the same as sonic beacon system, but easy of sharpening laser beam introduces accurate positional data. Fig.9 shows illustration of this beacon system.

5) Corner-cube Setting System⁽¹⁴⁾⁽¹⁵⁾

Three corner cubes are set on the way and vehicle has laser scanner with precision rotary encoder, vehicle can compute the position and heading data by on board computer. These data are introduced by use of geometrical relations also. The other method using two corner cubes and four laser transmitter/receiver and without scanner are also proposed in last September.

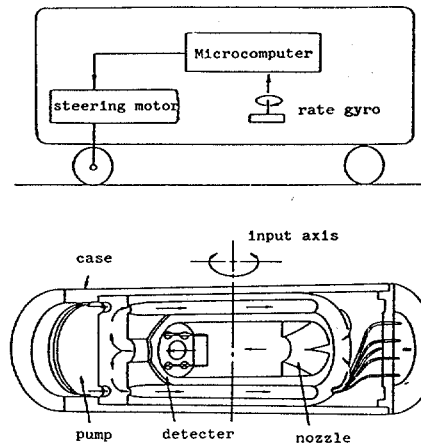


Fig. 8.

Self-sustained Guidance System using Rate Sensor with Rate Sensor Configuration

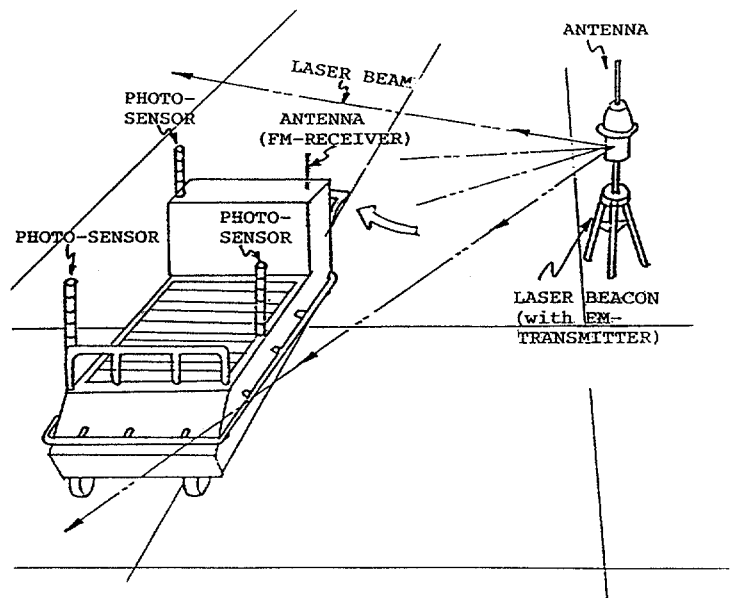


Fig. 9.

Illustration of Position Findings by Laser Beacon System

6) Recognition of Slit Pattern on Wall

On side wall of guide way, some slit patterns are set or written, vehicle can read the pattern optically by using the TV camera or something to find angular data. These data are applied to on board computer to find the positional data quickly. Fig.10 shows the measuring principle.

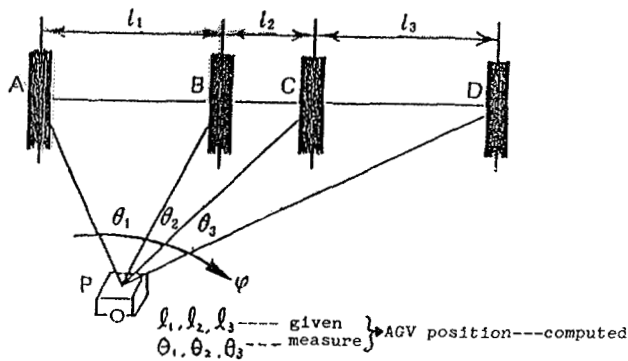


Fig. 10.
Principle of Position Findings by Slitt Pattern

COLLISION AVOIDANCE

Proximity sensor and touch sensor are used for almost all the AGV, and some recent developed AGV have good collision avoidance system using ultrasonic sensor, optical range sensor and video-image sensor, and so. Recently developed collision avoidance system consists of infrared LED transmitter and two sets of reflected light receivers. Lenz system with position sensing devices are good to assemble with reasonable costs. Measurement accuracy is about $1/100$ mm, and measurement angle is within 20 degrees. These positional information is used for avoidance obstacle on the route of AGV. If some obstacle is found, vehicle is controlled to avoid them by information of predetermined controlled to avoid algorithms, and nothing is found in front of the vehicle, preselected imaginary route is used for command signal of this vehicle. On these collision avoidance system, algorithms how to recover to the original route is important.

RECENT TOPICS ON AGV IN JAPAN

A recent discussion on advanced AGV and its system are presented on newspaper. By the paper, recent engineering point focused to the following theme:

- 1) flexibility of course setting,
- 2) dustless system,
- 3) low noise vehicle,
- 4) energy problem(battery problem)
- 5) self sustained guidance,
- 6) hybrid mode guidance(mixing self sustained with forced guide)
- 7) vehicle to vehicle communication,
- 8) system control or traffic control under the condition of many vehicle moves independently.
- 9) man-machine interface,
- 10) lower the cost, higher the performance,
- 11) transition of today's system to the new

- customers needs, customers such as agriculture, earth moving, building construction, office use etc.,
- 12) treaties in clean room like semiconductor and LSI factory,
- 13) safety problems.

CONCLUDING REMARKS

There are many engineering problems left for us. Automated guided vehicle system is very interesting object for mechanical, electrical, system and instrument engineers as well as computer engineering, and all must be considered the economical conditions and constraint. Here, new method for overcome many problems for AGV such as automatic guidance and automatic positionings are presented. But it should be said that this presentation is restricted only in Japan.

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